

ADARSH GAURAV

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PROFESSIONAL SUMMARY

Data Scientist with strong expertise in architecting scalable **data pipelines**, statistical predictive modeling, and production machine learning systems. Proven success designing end-to-end data workflows for real-world AI model training—improving Agentic AI task performance by **92%** and substantially reducing data processing latency across internal tools and multi-client projects. Skilled in exploratory data analysis (EDA), statistical inference, high-dimensional feature engineering, Bayesian optimization (Optuna), and SQL feature store pipelines. Proficient in Python, SQL, Scikit-learn, and PyTorch, with a track record of translating complex raw data into quantifiable business impact and low-latency production data solutions.

TECHNICAL SKILLS

Data Science & Modeling:	Supervised & Unsupervised Learning, Predictive Analytics, Statistical Inference & Hypothesis Testing, Exploratory Data Analysis (EDA), XGBoost, LightGBM, Random Forest, Logistic Regression, Cross-Validation, Optuna (Bayesian Tuning), Model Evaluation (ROC-AUC, PR-AUC, F1, RMSE)
Data Pipelines & Engineering:	Scalable ETL/ELT Pipelines, Feature Stores & Engineering, Data Ingestion & Cleansing, Class Imbalance Handling (SMOTE-NC), Time-Series & Signal Processing, SQL (PostgreSQL, MySQL - Advanced Window Functions, CTEs, Indexing), Data Drift Monitoring
Deep Learning & Agentic AI:	PyTorch, Scikit-learn, CNN, BiLSTM, Transformers, Multimodal Data Fusion, Agentic AI Workflows, Tool-Calling Validation, Retrieval-Augmented Generation (RAG), Vector Embeddings (BGE-M3), Semantic Search, OpenAI Whisper
Languages, DB & Tools:	Python (Advanced - Pandas, NumPy, Scipy), SQL, FastAPI, Git, GitHub, Docker (Basics), Google Cloud Platform (GCP), RESTful APIs, Matplotlib, Seaborn, Linux/Bash, VS Code, Google Colab

WORK EXPERIENCE

Data Science Intern | Darexai private limited **June 2026 – Sept 2026**

- Architected, built, and optimized end-to-end **data pipelines and ETL workflows** to ingest, clean, and preprocess high-throughput multimodal and structured datasets for training real-world **Agentic AI** models.
- Enhanced the execution accuracy and multi-step reasoning performance of **Agentic AI systems by 92%** through systematic prompt data curation, tool-calling validation schemas, and automated evaluation feedback loops.
- Streamlined real-time data streaming, feature extraction, and vector retrieval pipelines, significantly reducing system execution latency across existing internal developer tools and client project pipelines.
- Collaborated across **multiple concurrent client projects and internal data tools**, translating business requirements into scalable feature transformations, monitoring model drift, and ensuring high data fidelity via **Python and SQL**.

SELECTED DATA SCIENCE & AI PROJECTS

Multimodal Deep Learning System for Clinical Disease Prediction **PyTorch, Scikit-learn, FastAPI, PostgreSQL**

- Architected an end-to-end multimodal deep learning pipeline fusing continuous 12-lead ECG time-series signals with structured clinical laboratory records for early cardiovascular disease detection.
- Designed a hybrid **CNN-BiLSTM-Transformer** network with an attention-weighted feature bottleneck to jointly extract spatial waveform features and non-linear tabular interactions, beating unimodal tabular baselines by **15.6%**.
- Achieved **96.2% classification accuracy** and a **0.995 ROC-AUC** evaluated across 5-fold stratified cross-validation.
- Containerized the inference pipeline behind a high-throughput **FastAPI** REST service with PostgreSQL persistence, achieving sub-**150ms real-time latency**.

Autonomous RAG & Semantic Analytics Engine **Python, BGE-M3, Llama 3.2, Whisper, Scikit-learn**

- Engineered an automated **Retrieval-Augmented Generation (RAG)** analytics engine to transcribe, index, and query multi-hour unstructured audio/video lectures and technical corpora without external cloud API dependencies.
- Automated speech-to-text pipeline using **OpenAI Whisper (large-v2)** and FFmpeg audio stream slicing, generating structured, time-stamped text chunks for semantic indexing.
- Built a dense vector retrieval index with 1024-dimensional **BGE-M3 embeddings** and cosine similarity search (Scikit-learn/NumPy), achieving sub-**45ms query response times**.
- Integrated local LLM inference via **Ollama (Llama 3.2)** with custom hallucination-mitigation prompt templates, ensuring 100% of responses include grounded, exact temporal video citations.

Customer Churn & Lifetime Value (LTV) Predictive Pipeline **Python, XGBoost, LightGBM, SQL, Optuna, SHAP**

- Developed an end-to-end churn prediction pipeline on 250K+ customer profiles, engineering **30+ behavioral and Recency-Frequency-Monetary (RFM)** features using complex SQL window functions and Pandas.
- Trained and tuned **XGBoost and LightGBM** classifiers; mitigated severe class imbalance via **SMOTE-NC** and tuned hyperparameters with Bayesian optimization (**Optuna**), achieving **0.91 PR-AUC** and **86.5% Recall**.
- Leveraged **SHAP (SHapley Additive exPlanations)** for global and individual feature attribution, delivering interpretable churn risk factors to business stakeholders to proactively retain high-value accounts.

EDUCATION

Guru Ghasidas Vishwavidyalaya (A Central University) | Bilaspur, India **Dec 2022 – May 2026**

Bachelor of Technology (B.Tech) in Computer Science and Engineering

Relevant Coursework: Machine Learning, Probability & Statistics, Data Structures & Algorithms, Database Management Systems (DBMS), Artificial Intelligence, Operating Systems, OOP in Java/C++

CERTIFICATIONS & LEADERSHIP

- Deloitte Data Analytics Job Simulation (Forage):** Completed corporate analytics engagement — executed data cleaning, exploratory data analysis (EDA), hypothesis testing, and designed executive data dashboards.
- Student Leader, Pregrad:** Directed campus tech initiatives and coordinated 15+ student teams across hackathons and software development sprints.